



How Does External Social Network Affect Business Model Innovation in the Context of Digital Transformation: a Moderated Mediation Model

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Abstract

In the era of digital economy, business model innovation (BMI) has become an important way for companies to achieve digital transformation and gain new competitive advantages, but there is still a lack of research on the impact mechanism of BMI. Based on the social network theory perspective, this paper explores the influence of external social network (ESN) on BMI and the mediating role of business model boundary-spanning search (BMBS) and the moderating role of slack resources (SR). Through the survey and data analysis of 201 Chinese companies, this paper found that ESN has a significant positive impact on BMI. In addition, BMBS played a partial mediating role between ESN and BMI. At the same time, SR positively moderates the relationship between BMBS and BMI and the mediating effect of BMBS. The findings enrich the study about the influencing mechanism of BMI and provide insights into the BMI practice of companies in the context of digital transformation.

Keywords External social network · Boundary-spanning search · Slack resources · Business model innovation

Introduction

With the rapid development and popularity of emerging digital technologies such as big data, cloud computing, and artificial intelligence, more and more companies regard digital transformation as a key agenda to remain competitive in the digital era

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(Kraus et al., 2022). However, digital transformation requires not only the rational application of digital technologies or the cultivation of data resources but also companies to disrupt the traditional business logic of value creation and value capture and form new business models that are adapted to the digital transformation context (Verhoef et al., 2021). Yet, few scholars have conducted systematic and in-depth research on the influencing factors and process mechanisms of business model innovation (BMI) in the context of digital transformation (Ciasullo & Lim, 2022), which is not conducive to the development of BMI theory research, and it is also difficult to provide a reference for digital transformation companies to create value and establish competitive advantages through BMI.

Throughout the existing literature, scholars have richly explored the factors influencing BMI in traditional contexts from both external and internal aspects of the company (Foss & Saebi, 2017; Zhang et al., 2019). In terms of external influences, researchers have found that evolving consumer needs (Spieth et al., 2014), competitive environment changes (Pucihar et al., 2019), policy regime shocks (Bocken & Geradts, 2020), and opportunities presented by new information and communication technologies (Bouwman et al., 2019) all drive companies to innovate their business model in varying degrees. Regarding internal influencing factors, scholars' studies have shown that the cognitive patterns of top management (Frankenberger & Sauer, 2019), the resource capacity of the organization (Mielcarek & Dymitrowski, 2022), shifts in corporate strategy (Cortimiglia et al., 2016), and specific types of product innovation (Nubholz et al., 2019) are among the prominent internal reasons that influence companies' BMI.

As a whole, scholars have explored the factors influencing BMI based on various theoretical perspectives, such as institutional theory, cognitive theory, or dynamic capability theory, which have beneficially contributed to our understanding of the antecedent variables of BMI. But existing studies have relatively neglected the relationship between the external social network (ESN) in which a company is located and BMI, especially in the context of digital transformation, where companies' business models are not only affected by the spillover effects of digital transformation of peer companies but also need to deal with the challenges of digital-driven reconfiguration of stakeholder relationships such as customers, suppliers, and governments, making ESN an important factor influencing companies' change and innovative business model. According to social network theory, social network relationships created by organizations have social utility and provide access to quality information and resources, resulting in greater opportunities to influence organizational output (Molina-Morales & Martinez-Fernandez, 2010). Digital transformation has largely enhanced the connections between companies and members of ESN, such as suppliers, investors, customers, government, universities, and research institutions (Brunetti et al., 2020), enabling ESN to inform the direction of BMI and facilitating companies to access heterogeneous resources that can promote BMI from members of ESN (Khan et al., 2020). Therefore, this paper argues that ESN may be an important influencing factor for companies' BMI in the context of digital transformation.

Nevertheless, the process mechanism of ESN influencing BMI is still not clear enough, and companies' contact with members of ESN does not automatically improve BMI. Based on innovation search theory, Zhang and Li (2010) argue that

contact with external parties can contribute positively to company innovation by expanding the search scope and reducing search costs. Business model boundary-spanning search (BMBS) is the process of searching for the business model that can be applied or learned from within or between industries by a company across organizational boundaries and may be an effective way to drive BMI for the company (McDonald & Eisenhardt, 2020; Stott et al., 2016). Therefore, this paper introduces BMBS as a mediating variable when studying the direct effect of ESN on BMI, so as to further clarify the process mechanism of BMI.

Although it can enhance the BMI ability of companies, BMBS can also bring the following problems: as the scope of search expands, companies often have to pay higher search costs and invest resources to apply the search results. Slack resources (SR) help address the above problems. According to the resource-based view (RBV), SR makes companies more capable of trying out various options for action; thus, allowing them to experiment and take on risky activities (Mousa & Chowdhury, 2014; Soetanto & Jack, 2018). Specifically, companies with more SR are better equipped with the resources and capability base to conduct BMBS and are also better able to utilize the information and knowledge resources obtained via BMBS and integrate them into BMI. Therefore, we intend to incorporate SR into the research framework to explore the moderating role it plays between BMBS and BMI.

In summary, this paper focuses on BMI in the context of digital transformation from the ESN perspective and is primarily concerned with the following two research questions. First, based on social network theory and innovation search theory, we will explore the influence effect and process mechanism of ESN on BMI, and what role does BMBS play in the process? Second, based on the RBV, we verify whether SR as an organizational internal contextual factor will play a significant moderating effect in the above action mechanism. Through a combination of research methods such as multi-level regression analysis and Bootstrap, we attempt to reveal the action mechanism and boundary conditions of ESN on BMI to enrich the research in the field of BMI.

Theories and Hypotheses

ESN and BMI

According to the research of Foss and Saebi (2017), BMI is “designed, novel, non-trivial changes to the key elements of a firm’s business model and/or the architecture linking these elements.” Digital transformation has exposed companies to more opportunities and challenges in technology adoption, product development, and customer relationships, making the BMI process more uncertain and companies facing high risks and costs when making BMI decisions and actions. Therefore, companies must obtain and deliver market information promptly and obtain sufficient resources support to achieve BMI successfully. ESN, as a relationship network formed between companies and external organizations or individuals (Dai et al., 2015), has become an important platform for companies to obtain various types of information and resources for BMI.

On the one hand, ESN is an important source of new information for companies, such as external business knowledge, market trend information, changes in customer needs, and professional insights (Engelen et al., 2016). The ESN contains a large amount of valuable explicit and implicit information for digital transformation companies. The company uses its position in the social network structure to build communication and exchange channels and can reach more stakeholders (Luu & Ngo, 2019), providing a reliable guarantee for companies to obtain market information conducive to BMI. In addition, companies disseminate positive information about BMI through ESN, which can effectively reduce information asymmetry and gain members' trust in the social network, thereby reducing the external resistance to BMI. On the other hand, the willingness and ability of companies to use external resources are indispensable conditions for successful BMI. The resource allocation effect of ESN can help companies obtain external resource support in a timely and accurate manner with less resource consumption (Jiang et al., 2018). The trust, common values, and implicit rules formed in the process of building ESN can help companies easily obtain the resources needed for corporate activities (Uzzi, 1999). Therefore, the stronger the ESN of a company, the faster and more abundant resource support the company can obtain; thus providing a reliable resource foundation for BMI. Based on the reasons above, the following hypothesis is postulated.

H1: ESN has a positive impact on BMI.

The Mediating Effect of BMBS

The concept of the boundary-spanning search was first proposed by Rosenkopf and Nerkar (2001), which refers to companies searching for heterogeneous knowledge beyond existing boundaries to solve innovation problems. In this paper, BMBS refers to the company's search, integration, and use of external business model knowledge and resources to improve their development advantages, and it is an active search behavior to develop and meet the potential needs of the market. This paper holds that BMBS plays a mediating role between ESN and BMI.

ESN can promote the BMBS of companies. First, according to the social network theory, ESN provides a channel for companies to search for knowledge and imitate successful strategies across borders (Barroso-Castro et al., 2016). Companies can use ESN to expand the scope of their activities, easily access the business model of relevant network members through ESN, and help companies realize boundary-spanning search of business models. Second, ESN allows heterogeneous information to flow between companies (Lee et al., 2015; Wu et al., 2020), enabling companies to deeply understand the business model of other network members from different perspectives, so as to stimulate companies' willingness to boundary-spanning search. Third, ESN as a link between companies and external partners can effectively enable companies to discover valuable business models through network relationships (Gay, 2014). Moreover, close interaction with external network members enables companies to improve the accuracy and efficiency of boundary-spanning search by evaluating the matching and feasibility of these members' business

models, thereby enhancing the company's boundary-spanning search capabilities for the business model.

Furthermore, BMBS can drive companies to change their business model. Innovation search theory indicates that companies can optimize and improve their knowledge base through boundary-spanning search and better handle and solve problems in the innovation process (Ganco et al., 2020). First, boundary-spanning search behavior promotes the diversification of company information sources and the exchange of knowledge of different business models (Yan et al., 2020), increases the possibility of using an external business model to restructure the company business model, and provides an ideological basis for BMI. Second, in the process of BMBS, companies can strengthen their communication with external partners (such as suppliers, scientific research institutions, and consumers), which will help deepen the understanding of customer preferences and grasp the changes in market demand, so as to aim at the frontier of the market to achieve efficient BMI (Casadesus-Masanell & Zhu, 2013). Third, from the perspective of the spillover of the business model (Teece, 2010), the active BMBS can enable companies to deeply explore, make use of the advantages, and avoid the defects of the external business model through the observation and learning of the companies that have implemented specific business model, which helps break through the bottleneck of BMI by learning from the external business model.

Generally speaking, ESN has expanded the scope of actions and learning of the company, enabling the digital transformation company to communicate and cooperate with network members more easily, and has the opportunity to evaluate and observe their business model, which is conducive to the company's BMBS. In the process of BMBS, the company has broken the original business model concept and promoted BMI by learning from and using excellent external business models. Therefore, ESN can influence BMI through BMBS. Thus, this study assumes:

H2: BMBS mediates the link between ESN and BMI.

The Moderating Effect of SR

Slack is defined as “the difference between total resources and total necessary payments” (Cyert & March, 1963). According to RBV, as an important resource that can be deployed, SR can enable the company to strengthen control of internal activities, support company to carrying out boundary-spanning search and other activities, and promote the company to boldly try new opportunities (Daniel et al., 2004). Since digital transformation places higher resource requirements on company activities, this paper argues that SR can positively moderate the relationship between BMBS and BMI.

On the one hand, SR makes companies more flexible in their operations. They are not restricted by current market competition and can search for information and opportunities outside the market more freely (Bentley & Kehoe, 2020; Yang & Chen, 2017), which can enhance the ability of companies to conduct BMBS. In business practice, companies not only need to continue to pay attention to and

improve the existing business model that creates profits for them to strengthen their development capabilities but they also need to transcend business boundaries and identify those business models that can enter new strategic business areas, which is helpful to realize BMI and promote the rapid development of companies (Casadesus-Masanell & Tarzijan, 2012). In the above process, a large number of SR provides key support for BMI. On the other hand, SR provides an important impetus for companies to conduct BMBS to achieve BMI. When companies have more SR, driven by the increased cost pressure caused by idle resources, companies must try to find a way out for ready-made resources, and the important way of this process is to embed their resources into greater social needs, that is, create value through new business model (Amit & Zott, 2012; Carnes et al., 2019). Therefore, SR will drive companies to actively boundary-spanning search for business models and conduct BMI and use external business models as carriers to create new value from idle resources. Thus, the following hypothesis is proposed in this paper:

H3: SR has a moderating effect on BMBS and BMI.

SR provides resource support and search power for companies' BMBS. With the increase of SR, the ability and efficiency of BMBS will gradually improve, making companies more likely to develop BMI based on searching external business models. ESN provides convenient channels and opportunities for companies' BMBS, which is conducive to promoting BMI. Therefore, this paper believes that SR moderates the mediating effect of BMBS between ESN and BMI, that is, compared with low SR, ESN is more able to promote BMBS and then BMI in the context of high SR. Hence, this paper proposes the following hypothesis:

H4: SR moderates the mediating effect of BMBS on the relationship between ESN and BMI.

The conceptual framework of this study is shown in Fig. 1.

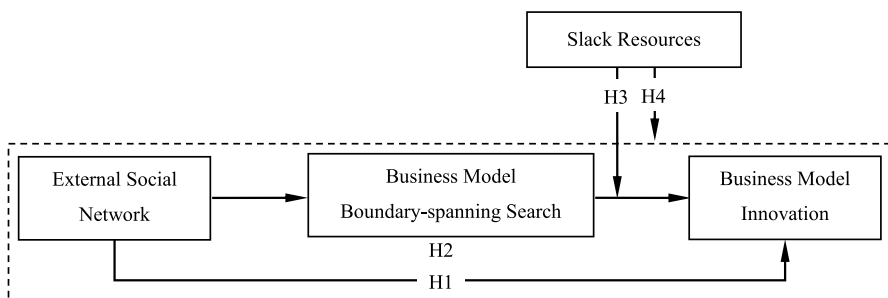


Fig. 1 Conceptual framework

Methodology

The objective of this paper is to examine the impact of ESN on BMI, as well as the mediating effect of BMBS and the moderating effect of SR; thus, the quantitative research approach is appropriate. In order to obtain the data needed for the study, we chose a sample of Chinese digital transformation companies to complete the data collection by designing and distributing a questionnaire. Then, based on the reliability and validity tests of the questionnaire, the research model was tested using hierarchical regression analysis and the Bootstrap method.

Survey and Data Collection

The effective matching of digital technology and business model is an important way for digital transformation companies to establish core competitive advantages. In practice, many digital transformation companies try to achieve further development in the digital age through BMI. Therefore, this paper chooses China's digital transformation companies as the research sample. Since the research variables of this paper focus on the company level, higher requirements are put forward for the questionnaire filling. This paper adheres to the following three criteria when selecting samples: first, the company is undergoing BMI or has a complete BMI experience; second, senior managers are required to fill in the questionnaire, because they can more objectively and accurately describe the relevant problems at the company level.

This paper takes China's digital transformation companies as a research sample and involves many industries such as intelligent equipment manufacturing, electronic information technology, industrial design, polymer materials, and biomedicine. In order to ensure the applicability of the questionnaire items, we randomly selected 15 companies for pre-survey (the 15 companies were not included in the formal survey), modified the questionnaire again according to the respondents' feedback, and finally formed the questionnaire officially used in this paper. The survey started in March 2021 and ended in May 2021. After communicating with executives by phone and email to solicit their willingness to participate in the survey, questionnaires were distributed to 486 companies. The final number of valid questionnaires was 201, and the effective recovery rate was 41.4%, which met the research needs of this paper. In terms of sample structure, the majority of industry types are intelligent equipment manufacturing, electronic information technology, and biomedical companies (60.2%), 54.7% of companies have fewer than 500 employees, and companies established within 10 years accounted for 58.2%. As for personnel characteristics of respondents, men accounted for 57.7%, those aged between 30 and 50 accounted for 70.1%, those with bachelor's degree or above accounted for 83.6%, and those with tenure of more than 5 years accounted for 68.2%. The statistical results of the basic information of the sample are shown in Table 1.

At the same time, in order to test the possible non-response bias in the sample data, this study divided all questionnaires into two groups according to the time sequence of collecting the questionnaires and performed an independent sample

Table 1 Basic information of the sample

Description	Frequency	Percentage (%)	Description	Frequency	Percentage (%)
Gender			Company age		
Male	116	57.7	Less than 5 years	42	20.9
Female	85	42.3	5–10 years	75	37.3
Personnel age			More than 10 years	84	41.8
Less than 30 years	37	18.4	Company size		
30–40 years	52	25.9	Less than 200 employees	38	18.9
41–50 years	89	44.3	200–500 employees	72	35.8
More than 50 years	23	11.4	More than 500 employees	91	45.3
Education			Industry type		
Specialist	33	16.4	Intelligent equipment manufacturing	34	16.9
Bachelor	97	48.3	Electronic information technology	19	9.5
Master	42	20.9	Industrial design	16	8.0
PhD	29	14.4	Polymeric materials	29	14.4
Tenure			Biomedicine	23	11.4
1–5 years	64	31.8	Garment manufacturing	22	10.9
6–10 years	85	42.3	Services	17	8.5
More than 11 years	52	25.9	Others	41	20.4

T-test. The results showed that the *t* values of gender, age, education, and tenure were 0.165 ($p=0.269>0.05$), 0.187 ($p=0.852>0.05$), -0.118 ($p=0.906>0.05$), and 0.137 ($p=0.891>0.05$), respectively. Therefore, there is no serious non-response bias in this study.

Measurements

The scales used in this paper are all derived from mature scales in existing studies. To improve the applicability of the scales, the reference scales were appropriately translated and back-translated in the process of designing the scales, and the question items were modified according to the characteristics of the study. We used a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree) to measure the multi-item constructs.

External social network (ESN): Based on the research of Cuevas-Rodriguez et al. (2014), three items were designed. A sample item was “The company and partner companies have highly identical goals and interests in common projects.” Cronbach’s α value for the scale was 0.713.

Business model boundary-spanning search (BMBS): Integrating studies drawing on Sidhu et al. (2007) and Layrisse et al. (2020), BMBS was measured by four items. A sample item was “The company can discover new business model

in new markets and new customer groups.” Cronbach’s α value for the scale was 0.741.

Slack resources (SR): Drawing on the research of Tan and Peng (2003), six items were designed to measure SR. A sample item was “The company has sufficient financial resources to meet sudden investment needs.” Cronbach’s α value for the scale was 0.719.

Business model innovation (BMI): We measured BMI using four items from Zott and Amit (2008). A sample item was “The company can conduct business model innovation to meet the new needs of customers.” Cronbach’s α value for the scale was 0.812. Table 2 lists the measurement items.

Control variables: Referring to the research of Basic (2022), this paper used company age, company size, and industry type as control variables. The longer a company has been operating in the market, the greater the organizational inertia and the lower the willingness to change and innovate in the organization; hence, it is necessary to control company age. Company age is measured as the natural logarithm of the company’s establishment time. Company size is an important indicator of a company’s market competitiveness and affects the resources devoted to innovation activities. The natural logarithm of the number of employees in a company is used to measure company size. The BMI of the high-tech company may be different from other companies; thus, this paper controls whether the company is a high-tech company or not. The industry type is measured using a dichotomous variable, with 0 being a non-high-tech company and 1 being a high-tech company.

Results

Validity and Common Method Variance Test

The average variance extracted values of ESN, BMBS, SR, and BMI are 0.509, 0.535, 0.531, and 0.554, respectively, which are greater than 0.50. Therefore, these variables possess strong convergent validity. Discriminant validity was tested using confirmatory factor analysis. Compared with other models (see Table 3), the fit index of the four factors model is the best ($\chi^2/df=1.500$; CFI=0.954; TLI=0.944; IFI=0.955; RMSEA=0.050), indicating that the four latent variables in this paper have good discriminant validity. In addition, the square root of average variance extracted of variables is greater than correlations with the dimensions between different variables (see Table 4), implying that the measurements have adequate discriminant validity.

We tested the possible common method variance by adding a common method factor. As shown in Table 3, after adding the common method factor, all indicators of the model have not changed significantly, indicating that the problem of common method variance in this study is not serious and will not affect the effectiveness of the research results.

Table 2 Measurement items

Construct	Item	Factor loading	AVE	Cronbach's α	Composite reliability
ESN	ESN 1: Our company and partner companies have highly identical goals and interests in common projects	0.712	0.509	0.713	0.756
	ESN 2: Our company and partner companies trust each other and often work together to develop new and improve existing products	0.730			
	ESN 3: Partner companies share a common vision with our company in terms of market environment and achievement success	0.697			
BMBS	BMBS 1: Our company can discover new business model in new markets and new customer groups	0.761	0.535	0.741	0.821
	BMBS 2: We keep an eye on other industries that are related to our company in terms of business model	0.703			
	BMBS 3: Our company closely tracks the business model of companies that serve the same target customers as we do	0.718			
SR	BMBS 4: Our company pays close attention to the business model of peer companies that are not located in the same region	0.742	0.531	0.719	0.871
	SR 1: Our company has sufficient financial resources to meet sudden investment needs	0.642			
	SR 2: Our company has enough retained surplus to cover our market development needs	0.664			
	SR 3: Our financial resources are flexible according to the situation	0.823			
	SR 4: Our company's equipment and facilities are often not operating at full capacity	0.772			
	SR 5: There are still many underutilized resources in our operation	0.770			
BMI	SR 6: Our company has surplus production (service) capacity, and there is still more room to play	0.683	0.554	0.812	0.832
	BMI 1: Our company can conduct business model innovation to meet the new needs of customers	0.777			
	BMI 2: Our company's business model uses an innovative trading approach	0.764			
	BMI 3: Our company's business model creates new ways to profit	0.731			
	BMI 4: Our company's business model is novel	0.703			

AVE, average variance extracted; ESN, external social network; BMBS, business model boundary-spanning search; SR, slack resources; BMI, business model innovation

Table 3 Validity test

Model	χ^2	df	χ^2/df	CFI	TLI	IFI	RMSEA
One factor	416.530	118	3.530	0.753	0.715	0.756	0.112
Two factors	281.987	117	2.410	0.863	0.841	0.865	0.084
Three factors	174.177	115	1.515	0.951	0.942	0.952	0.051
Four factors	167.982	112	1.500	0.954	0.944	0.955	0.050
Four factors + common method factor	125.435	97	1.293	0.976	0.967	0.977	0.038

Four factors: ESN, BMCS, SR, and BMI; three factors: ESN+BMCS, SR, and BMI; two factors: ESN+BMCS+SR and BMI; one factor: ESN+BMCS+SR+BMI

Descriptive Statistics

Table 4 presents the means and standard deviations for each variable and inter-variable correlations for all study variables.

Hypotheses Testing

Mediating Effect Testing of BMBS

This paper uses hierarchical regression analysis to test the mediating effect of BMBS. The results in Table 5 show that after adding BMBS, the impact of ESN on BMI decreases from 0.363 to 0.208. Therefore, BMBS plays a partial mediating role between ESN and BMI.

This paper further uses the process plug-in program of SPSS to test the mediating effect with Bootstrap = 5000 and 95% confidence interval as the standard. It can be seen from Table 6 that the direct effect of ESN on BMI is significant ($\beta = 0.240$, 95% CI (0.043, 0.436)). BMBS has a significant mediating effect between s ESN and BMI ($\beta = 0.177$, 95% CI (0.037, 0.310)). Combining the hierarchical regression analysis and Bootstrap estimation results, Hypothesis 1 and Hypothesis 2 are verified.

Table 4 Descriptive statistics and correlations for variables

Variables	Mean	SD	ESN	BMBS	SR	BMI
ESN	4.002	0.630	0.713			
BMBS	4.015	0.566	0.656***	0.731		
SR	3.750	0.569	0.420*	0.508***	0.729	
BMI	3.755	0.724	0.383***	0.392***	0.583***	0.744

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; numbers in bold on the diagonal indicate the square root of average variance extracted

Table 5 Results of hierarchical regression analysis

Variables	BMBS	BMI	
	Model 1	Model 2	Model 3
Control variables			
Company age	-0.006	-0.018	-0.017
Company size	0.072	0.115*	0.097
Industry type	-0.045	0.006	0.016
Independent variable			
ESN	0.653***	0.363***	0.208*
Mediating variable			
BMBS			0.236**
R ²	0.437	0.160	0.191
F	38.014***	9.316***	9.217***

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$ **Table 6** Bootstrap estimation of the mediation model

Path	Coefficients	95%CI	
		LLCI	ULCI
ESN → BMI	0.240	0.043	0.436
ESN → BMBS → BMI	0.177	0.037	0.310

Moderating Effect Testing of SR

This paper uses the Bootstrap method to test the moderating effect of SR, and the results are shown in Table 7. The F value of the model is 19.661 ($p = 0.000 < 0.001$), reaching a significant level. The R^2 value of the model is 0.378, indicating that the overall interpretation ability of the model reaches 37.8%.

It can be seen from Table 7 that the regression coefficient of BMBS on BMI is 0.181, and the 95% CI is (0.010, 0.350); the regression coefficient of SR on BMI is 0.660, and the 95% CI is (0.492, 0.828); the regression coefficient of the interaction term of BMBS and SR on BMI is 0.308, and the 95% CI is (0.042, 0.574), indicating that SR plays a moderating role in the relationship between BMBS and BMI. At the same time, it can be seen from Fig. 2 that compared with low-SR, BMBS under high-SR has a stronger impact on BMI, so Hypothesis 3 has been verified.

The Bootstrap method tests the moderating effect of SR in the mediating relationship. It can be seen from Table 8 that SR plays a significant positive moderating role in the mediating path “ESN → BMBS → BMI,” and Hypothesis 4 is verified.

Table 7 The moderating effect of SR

	Coefficient	SD	95% CI		p
			LLCI	ULCI	
Control variables					
Company age	0.005	0.057	-0.111	0.112	0.993
Company size	0.040*	0.022	0.004	0.094	0.046
Industry type	-0.075	0.087	-0.247	0.095	0.384
Independent variable					
BMBS	0.181*	0.086	0.010	0.350	0.037
Moderating variable					
SR	0.660***	0.085	0.492	0.828	0.000
Interaction term					
BMBS*SR	0.308*	0.135	0.042	0.574	0.024
R ²	0.378				
F	19.661***				

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

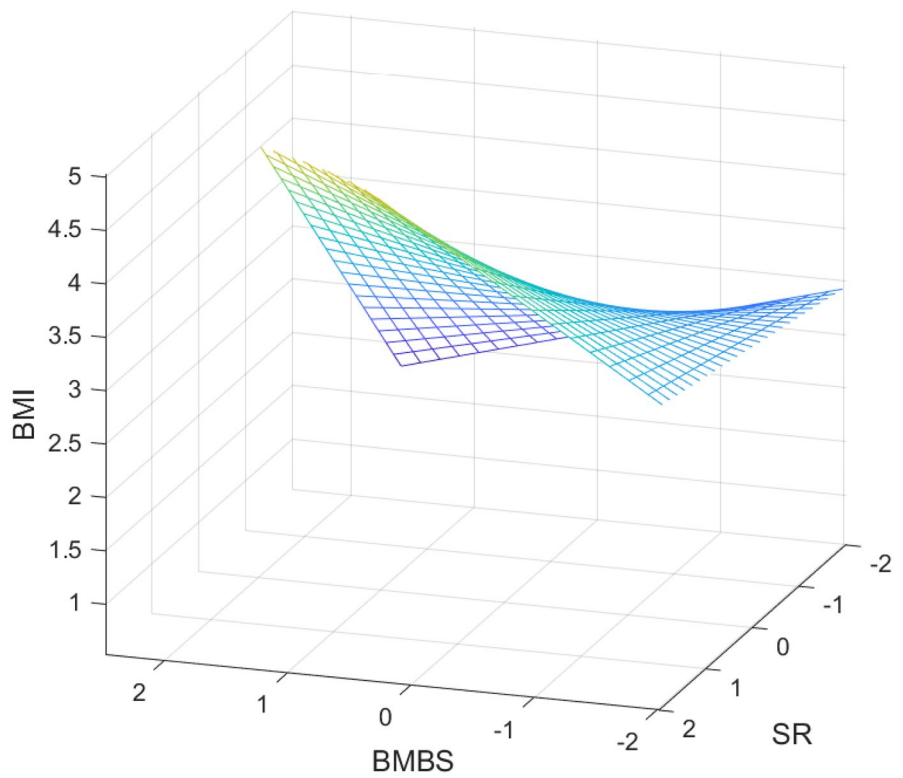


Fig. 2 The moderating effect of SR

Table 8 Moderated mediation result of SR

Index	BootSE	BootLLCI	BootULCI
0.212	0.088	0.040	0.388

Discussion and Conclusions

In this paper, using BMBS as a mediating variable and SR as a moderating variable, we explore the impact mechanism of ESN affecting BMI in the context of digital transformation and obtain the following three conclusions.

First, ESN has a significant positive impact on BMI. ESN enhances communication and collaboration with external stakeholders, enabling companies to access more business information and resource support (Leenders & Dolfisma, 2016; Thompson, 2018), which provides favorable conditions for companies to conduct BMI. Therefore, when a company has a powerful ESN, it has a better possibility and ability to achieve BMI. Although scholars have previously explored the impact of ESN on company innovation and strategic change (Cuevas-Rodriguez et al., 2014; Riviezzo & Garofano, 2018), they were mostly based on traditional development contexts, while in the context of digital transformation, companies face a more challenging and uncertain business environment, which makes the relationship between ESN and BMI a new issue worth exploring. Through in-depth analytical argumentation and empirical testing, this paper reveals how ESN drive companies' BMI decisions and actions by providing information and resource support in the context of digital transformation, providing a useful addition to the research in the field of social networks and business model. At the same time, the findings of this paper support and validate the view of Nu'man et al. (2020) that social networks facilitate cooperation and information sharing among economic agents and are therefore essential for business activities.

Second, BMBS mediates the relationship between ESN and BMI. Specifically, ESN provides good channels and heterogeneous information for companies to search for business model across borders, which effectively enhance the ability of companies to conduct BMBS. Furthermore, in the process of BMBS, companies are exposed to different business models, and through the learning and utilization of external business models, they have more confidence and willingness to pursue BMI. The research of Schotter et al. (2017) showed that creative and competitive companies tend to use their ESN to overcome resistance to local search and cross organizational boundaries to explore thoroughly, which facilitates companies to achieve great innovation. And this paper finds that the abovementioned research views still have strong explanatory power for the current reality of BMI in the digital era. ESN inevitably influences the implementation and effect of BMBS, and this boundary-spanning search behavior will in turn affect the BMI of companies. The empirical results of this paper also confirm the findings of Kim et al. (2018) that boundary-spanning search has a top-down role in the process of internalizing external relational resources into company performance, suggesting that

BMBS is an effective way for companies to break out of business model inertia and acquire new knowledge and ideas from ESN and thus promote BMI.

Third, SR reinforces the direct effect of BMBS on BMI and the mediating effect of BMBS. Although Basu et al. (2007) view SR as a waste of organizational resources and can lead to lower productivity of companies, this paper finds that SR, as resources that can be freely used by companies, can have a positive effect and give companies more options when faced with opportunities. The results of the empirical study also indicate that under the condition of high SR, companies are more able to expand the scope and increase the effectiveness of BMBS. The study by Soetanto and Jack (2018) found that any form of SR can be adapted and utilized in some way; thus, giving the company a good resource base and risk buffer and contributing to good conditions for innovation. The results of this paper support these insights specifically that SR guarantees the basic resources needed for risk-taking activities and BMI, enabling companies to carry out BMI based on new market opportunities and searched external business models. This paper also responds to calls for a deeper exploration of the role of SR in organizational decisions and activities (Bentley & Kehoe, 2020). This paper considers SR as an important boundary condition in the relationship between ESN, BMBS, and BMI, which not only brings important insights to the research on SR but also provides a useful empirical test of the ability of SR to influence value creation and exert contextual effects.

Theoretical Implications

The research of this paper has some theoretical implications. First, this study promotes the development of social network theory. Although previous studies have emphasized the importance of ESN for company transformation, few studies have systematically studied ESN and BMI. Considering the important role of BMI in company development, this paper investigates whether ESN can promote BMI in the context of digital transformation and fills the gap in the research. Second, this study contributes to the development of innovation search theory. This paper points out that ESN is an important driver of BMBS, and ESN provides channels and convenience for BMBS, which is conducive to promoting the research on the influencing factors of innovation search in the field of innovation search theory. Finally, existing research is still in a theoretical black box regarding the process mechanism of ESN influencing BMI. This paper responds to this question by combining social network theory, innovation search theory, and RBV, revealing and confirming that BMBS is an important process mechanism through which ESN impacts BMI, and that SR plays an important moderation role in this relationship.

Managerial Implications

The findings of this paper also have some practical implications. First, ESN can provide important external support for companies to realize BMI. Therefore, in the process of daily operation, companies should strengthen communication and contact with external network members, such as suppliers, R&D organizations, government,

and customers, and establish long-term mutually beneficial cooperative relationships with each other; thus, improving their ability to obtain important information and business resources from external networks. Second, companies should consciously cultivate and improve their capabilities of BMBS, have the courage to break industry boundaries and expand their business horizons, take the initiative to seek, observe and learn from excellent business models from other industries and companies, and use BMBS to provide new ideas and new thinking for companies, thereby promoting the implementation and achievement of BMI. Third, it is necessary for companies to acquire and maintain a certain amount of SR in their business process, which can be used to counteract uncertainty and explore new opportunities. In the context of digital transformation, companies face greater challenges. They should pay more attention to devoting SR to BMI activities to achieve the conversion of resources to business performance and competitive advantage.

Limitations and Future Research

This paper also has some limitations, which provide research directions for future studies. First, this paper is mainly based on cross-sectional data, which weakens the reliability of the causal relationships between variables. Future studies could use longitudinal data and lag the BMI by several periods to enhance the persuasiveness of the causal relationships. Second, this paper only explores the moderating role of SR and does not classify SR. Due to the fact that various types of SR may have diverse impacts on companies, future research can distinguish different SR, such as financial slack and human slack.

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Declarations

Competing Interests The authors declare no competing interests.

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